FFT Self-Gravity in the AthenaK Shearing Box: Implementation and Validation

`athenak-multigrid` tree, branch `multigrid`

August 3, 2026

Abstract

This document records the design, mathematics, implementation, and validation of an FFT-based self-gravity solver for AthenaK's shearing box, added to the `athenak-multigrid` tree. The solver follows the Athena-C implementation in `selfg_fft_disk.c`: the shearing-box phase-shift method of Gammie (2001) and the open (vacuum) vertical boundary condition of Koyama & Ostriker (2009), built on the `kokkos-fft` library so the same code runs on CPU and GPU backends. Validation: the triply periodic solve satisfies the discrete Poisson equation to machine precision (3.8×10^{-14}); the shearing solve is exact at shear-periodic instants and converges at the remap interpolation order (3rd for `ppmx`) against analytic shearing-wave solutions; the open-boundary solve is exact in the interior and matches the analytic slab potential to 0.15%; a Jeans-wave evolution agrees with the existing multigrid solver to nine significant digits while running 3.5× faster; and a self-gravitating shearing wave reproduces linear swing-amplification theory through the swing (peak amplification 3.62 vs. 3.605 predicted at 256^2), with the wave's phase locking to the instantaneous shearing wavevector holding to 10^{-13} and the remaining L_2 difference shown to be hydrodynamic in origin — it is the same size in a control run with gravity switched off.

Contents

1 Plan and scope

1.1 Goal and roadmap

The scientific target is shearing-box simulations combining gas self-gravity, dust with mutual drag (streaming instability, including the background radial pressure gradient), and mesh refinement that zooms onto high dust-density regions in spiral arms launched by gravitational instability (cf. Baehr, Zhu & Yang 2022, ApJ 933, 100, who omitted the pressure gradient). Because refinement is essential, the adopted route makes the *multigrid* solver shear-aware, with the FFT solver of this document retained in three roles: the machine-precision *oracle* for validating multigrid's shear BCs, the production solver for uniform-grid runs, and the donor of exact open-z Dirichlet boundary values for stratified multigrid runs. The phases:

1. **(done; Secs. 2–5)** Uniform-grid FFT Poisson solver: shearing-periodic horizontal boundaries (Gammie 2001 phase shift), periodic or open vertical boundaries (Koyama & Ostriker 2009), validated through swing amplification.

2. **(done)** Phase 0: shearing box + mesh refinement infrastructure — interior-x_1 refinement policy, a non-FARGO mode, and orbital advection (FARGO) on refined meshes under an annular refinement policy. Documented separately in `multigrid_selfgravity_validation.pdf`, which covers the multigrid/refinement track.

3. Phase 1–2: shear-periodic boundary conditions in the multigrid driver, validated against the FFT solver of this document as a machine-precision oracle; then multigrid gravity on refined shearing-box meshes.

4. Phase 3–5: merge with the `athenak-dust` particle-mesh dust module (dust density in the Poisson source, gravity on particles, pressure-gradient source term), stratification with open-z boundaries, and science runs.

1.2 Version-1 scope

The solver currently *requires* (and enforces with fatal errors): a 3D mesh, a uniform grid (no SMR/AMR — AthenaK already forbids refinement with a shearing box), a single MPI rank, and (shear-)periodic x_1/x_2 boundaries. The vertical boundary is periodic (with mean-density subtraction) or open. Shearing and non-shearing cases share one code path: with shear rate $q = 0$ every shear-specific step reduces to the identity.

2 Numerical method

2.1 The discrete Poisson equation and its exact k-space inverse

The gravity source terms difference the potential with second-order central differences, so the solver targets the *finite-difference* Poisson equation

$$(L\Phi)_{ijk} \;\equiv\; \frac{\Phi_{i+1} - 2\Phi_i + \Phi_{i-1}}{\Delta x^2} + \frac{\Phi_{j+1} - 2\Phi_j + \Phi_{j-1}}{\Delta y^2} + \frac{\Phi_{k+1} - 2\Phi_k + \Phi_{k-1}}{\Delta z^2} \;=\; 4\pi G\,\rho_{ijk}, \tag{1}$$

not the continuum equation. On a periodic uniform grid the DFT modes $e^{\mathrm{i}k_x x_i}$ are eigenvectors of each difference operator: applying the stencil to $e^{\mathrm{i}k_x i\Delta x}$ and factoring out the mode gives the eigenvalue

$$D(\mathbf{k}) = \frac{2\cos(k_x\Delta x) - 2}{\Delta x^2} + \frac{2\cos(k_y\Delta y) - 2}{\Delta y^2} + \frac{2\cos(k_z\Delta z) - 2}{\Delta z^2} \xrightarrow{k\Delta\to 0} -k^2. \tag{2}$$

The solve is therefore: FFT $4\pi G\rho$, divide each mode by $D(\mathbf{k})$, inverse FFT. Because D is the exact symbol of the same stencil later used to compute forces, the returned Φ satisfies Eq. (1) to machine precision — there is no iteration residual (contrast multigrid). The $\mathbf{k} = 0$ mode is set to zero, which is exactly equivalent to subtracting the mean density (the Jeans swindle, required for consistency of the periodic problem).

2.2 Shearing box: the phase-shift (rolled-frame) method

With background velocity $v_{y,0} = -q\Omega x$, the shearing-periodic boundary condition identifies

$$\rho(x,\, y,\, z,\, t) \;=\; \rho(x + L_x,\; y - q\Omega L_x t,\; z,\; t), \tag{3}$$

so the box is strictly periodic only at times $t_n = n\,T_{\rm sh}$ with $T_{\rm sh} = L_y/(q\Omega L_x)$. Let t_s be the time since the most recent t_n and $\theta \equiv q\Omega t_s$ (code variable `qomt`). In sheared coordinates $y' = y + \theta x$ the density *is* doubly periodic:

$$\rho'(x,\, y',\, z) = \rho(x,\; y' - \theta x,\; z) \qquad \Longleftrightarrow \qquad \text{“roll”: shift each } x\text{-column of content by } +\theta x \text{ in } y. \tag{4}$$

A rolled-frame mode $e^{\mathrm{i}(k'_x x + k_y y')}$ is a shearing wave with *physical* radial wavenumber

$$k_x = k'_x + \theta\, k_y \qquad \text{(index form: } k_x\Delta x = \big(i_p + \theta\,\tfrac{L_x}{L_y}\,j_p\big)\tfrac{2\pi}{N_x}\text{)}, \tag{5}$$

with j_p the *signed* mode index. The pipeline is: roll $\rho \to$ FFT $\to$ divide by $D(k'_x{+}\theta k_y,\, k_y,\, k_z) \to$ inverse FFT $\to$ unroll Φ (shift by $-\theta x$).

A point worth recording: if the roll/unroll used *spectral* (Fourier) interpolation, the composite operation would invert Eq. (1) exactly at all times — the per-column shift is then a pure phase factor $e^{\mathrm{i}k_y\theta x}$, under which the unsheared stencil pulls back exactly to the shifted-eigenvalue form of Eq. (5). All error at $\theta \neq 0$ therefore comes from the finite-order remap interpolation, and the solver is exact whenever $\theta = 0$ (verified below to 3×10^{-14}).

The remap is the conservative flux form shared with AthenaK's orbital advection (`remap_fluxes.hpp`): the integer part of the shift is folded into a periodic-wrapped load, the fractional part $\epsilon \in (-1, 1)$ is applied as $q_j \to q_j - (F_{j+1} - F_j)$ with upwind flux kernels of first (`dc`), second (`plm`), or third (`ppmx`) order. As a *static* remap, donor-cell reduces to linear interpolation (2nd-order pointwise), `plm` is 2nd order with a smaller constant, and `ppmx` is 3rd order — exactly the orders measured in Sec. 5.

Ghost zones. The gravity source term needs one valid face-adjacent ghost layer of Φ. Interior meshblock faces come from the global solution; x_2 ghosts wrap periodically; the physical x_1 ghost planes use the shear-periodic identification: the inner ghost plane is the outer boundary column with content shifted by $+\theta L_x$ (mod L_y), and vice versa, using the same remap kernel.

2.3 Open vertical boundary: the two-solve method (Koyama & Ostriker 2009)

Fourier transforming horizontally, each mode obeys $(\mathrm{d}^2/\mathrm{d}z^2 - k_\perp^2)\hat{\Phi} = 4\pi G\hat{\rho}$, whose isolated (vacuum) solution is the screened kernel $\hat{\Phi}(z) = -\frac{2\pi G}{k_\perp}\int \hat{\rho}(z')\, e^{-k_\perp|z-z'|}\mathrm{d}z'$. A vertically periodic solve instead delivers an infinite stack of image slabs separated by L_z:

$$S_p = \sum_{n=-\infty}^{\infty} e^{-k_\perp|z-z'-nL_z|} = \frac{e^{-k_\perp\zeta} + e^{-k_\perp(L_z-\zeta)}}{1-q}, \qquad q \equiv e^{-k_\perp L_z},\ \ \zeta = z - z' \in [0, L_z). \tag{6}$$

A second solve on $4\pi G\rho\, e^{-\mathrm{i}\pi z/L_z}$ shifts the vertical spectrum by half a wavenumber (half-integer harmonics = *antiperiodic* vertical extension), whose Green's function is the alternating stack

$$S_a = \sum_n (-1)^n e^{-k_\perp|z-z'-nL_z|} = \frac{e^{-k_\perp\zeta} - e^{-k_\perp(L_z-\zeta)}}{1+q}. \tag{7}$$

Both are combinations of the same two functions of ζ; solving the 2×2 system for the weights that cancel the wraparound term and normalize the vacuum kernel gives

$$\boxed{\alpha_p = \tfrac{1}{2}\,(1 - e^{-k_\perp L_z}), \qquad \alpha_a = \tfrac{1}{2}\,(1 + e^{-k_\perp L_z}), \qquad \alpha_p S_p + \alpha_a S_a = e^{-k_\perp\zeta}.} \tag{8}$$

Every periodic image cancels identically; the result is the isolated-slab potential at the cost of one extra FFT pair. In k-space, solve A multiplies by $\alpha_p/D(k'_x, k_y, k_z^{\mathrm{int}})$ and solve B by $\alpha_a/D(k'_x, k_y, k_z^{\mathrm{half}})$ with $k_z^{\mathrm{half}}\Delta z = (m + \frac{1}{2})\,2\pi/N_z$; the recombination is $\Phi = \Phi_A + \mathrm{Re}\big[e^{+\mathrm{i}\pi\tilde{z}/L_z}\Phi_B\big]$ with cell-centered $\tilde{z} = z - z_{\mathrm{min}}$.

At $k_\perp = 0$ the weights become $(0, 1)$: the horizontally averaged density is carried entirely by the antiperiodic solve, which correctly reproduces the linear $|z|$-type slab potential; no mean-density subtraction occurs in open mode, and the $D = 0$ singularity never arises. Matching Athena-C, the screening exponent uses the *continuum* $k_\perp = [(k_x\Delta x/\Delta x)^2 + (k_y\Delta y/\Delta y)^2]^{1/2}$ (with the shear-shifted k_x) rather than the discrete decay rate κ defined by $2[\cosh(\kappa\Delta z) - 1]/\Delta z^2 = k_{\perp,\mathrm{eff}}^2$; the difference matters only where $e^{-k_\perp L_z}$ is already exponentially negligible. The x_3 ghost layer is filled by linear extrapolation in open mode (the layers touching it are excluded from "interior" error norms).

2.4 Cost and normalization

Periodic mode: one complex-to-complex FFT pair per solve. Open mode: two pairs. FFT plans are created once and reused; normalization is the NumPy convention (inverse transform carries $1/N$), passed explicitly. Kernel coefficients (Eqs. 2, 5) are computed on the fly inside the k-space multiply — they are time-dependent through θ, and recomputing costs less than storing. The solve runs once per Runge–Kutta stage, matching the multigrid convention, so the potential is consistent with the stage's density.

3 What was built

3.1 New and modified files

File	Content
`src/gravity/fft_gravity.hpp`	`FFTGravitySolver` class declaration
`src/gravity/fft_gravity.cpp`	solver implementation (all kernels)
`src/gravity/gravity.{hpp,cpp}`	solver switch, `Gravity::Solve()` dispatch
`src/driver/driver.cpp`	one line: `pgrav->Solve(this,stage)`
`src/multigrid/multigrid.hpp`	pre-existing fix: `Smooth` moved out-of-line†
`src/pgen/tests/fft_poisson.cpp`	validation problem generator (+ registration)
`src/pgen/tests/swing.cpp`	swing-amplification pgen + user history function
`CMakeLists.txt`, `src/CMakeLists.txt`, `config.hpp.in`	build integration
`inputs/tests/*.athinput`	test inputs (fft_poisson*, jeans_fft, swing)
`validation/`	this document, figures, Julia notebook

† `Multigrid::Smooth` accessed the not-yet-defined `MultigridDriver` inside the class body; Apple clang 21 rejects this, so the template body was moved below the driver class definition. No behavior change; the multigrid Jeans test still passes.

3.2 Solver pipeline

`FFTGravitySolver::Solve()` executes, per RK stage:

1. **Gather**: active-zone density from all meshblocks into a global (N_z, N_y, N_x) device array (block offsets from logical locations), scaled by $4\pi G$.
2. **Roll**: conservative per-column y-shift by $+\theta x$ (Eq. 4); identity when $\theta = 0$.
3. **k-space solve**: periodic (one FFT pair) or open (two pairs, Sec. 2.3), with the shear-shifted kernel (Eq. 5).
4. **Unroll**: shift by $-\theta x$.
5. **Ghost planes**: shear-periodic x_1 ghost columns ($\pm\theta L_x$ shifts).
6. **Scatter**: into `pgrav->phi` (active zones + one ghost layer); the unchanged `SelfGravity` source term consumes it.

The multi-rank seam: steps 1 and 6 are the only places that assume the pack spans the mesh; an `MPI_Allgatherv` between local gather and roll (and a rank-local scatter) upgrades the solver without touching steps 2–5.

3.3 Runtime options

Block/parameter	Values	Default	Meaning
`<gravity> solver`	`multigrid` | `fft`	`multigrid`	Poisson solver selection
`<gravity> four_pi_G`	real	-1 (unset)	$4\pi G$ (pgens may override)
`<gravity> vert_bc`	`periodic` | `open`	`periodic`	vertical boundary
`<gravity> fft_remap`	`dc` | `plm` | `ppmx`	`plm`	y-remap order
`<shearing_box> qshear, omega0`	real	—	read if the block exists
`<hydro_srcterms> self_gravity`	bool	false	enable source term

Existing multigrid input files run unchanged (default `solver=multigrid`).

4 Building and compiling

Prerequisites (done once; already satisfied on this machine):

```
cd athenak-multigrid
git submodule update --init kokkos     # Kokkos 4.7.02, vendored
brew install fftw                      # host FFT backend for kokkos-fft
```

Configure and build (kokkos-fft v2.0.0 is fetched automatically by CMake; it consumes the in-tree Kokkos target and links FFTW on host backends, cuFFT/hipFFT on GPU backends):

```
mkdir build && cd build
cmake -D Athena_ENABLE_FFT=ON ..
make -j8
```

Without `-D Athena_ENABLE_FFT=ON` the code builds exactly as before; requesting `solver=fft` then aborts with an explanatory error.

5 Tests and results

All tests use the `fft_poisson` problem generator, which fills a deterministic density field, calls the gravity solve once, and prints error diagnostics (lines starting `# FFT-POISSON`). Initial conditions are selected by `<problem> profile = modes | shwave | slab` and are overridable on the command line. The companion Julia notebook (`validation/fft_selfgravity_tests.ipynb`) re-runs every test and checks the outputs independently of the C++ diagnostics.

5.1 Discrete consistency (periodic, no shear)

Multi-mode + Gaussian-blob density, 64^3, 8 meshblocks:

```
./build/src/athena -i inputs/tests/fft_poisson.athinput
# FFT-POISSON ERRORS: max_rel_all= 3.757e-14  l2_rel_all= 1.216e-14
```

The 7-point Laplacian of the returned potential reproduces $4\pi G(\rho-\bar{\rho})$ at machine precision, confirming the k-space inversion, the multi-block gather/scatter, and the periodic ghost fill simultaneously.

5.2 Shear bookkeeping: exactness at shear-periodic instants

Same field in a $q = 1.5$, $\Omega = 1$ shearing box ($T_{\rm sh} = 2/3$): solving at $t = T_{\rm sh}$ (i.e. $\theta = 0$) returns `max_rel = 3.0e-14` — the entire shear code path (roll, shifted kernel, unroll, shear ghosts) collapses to the exact periodic solve, as it must. At a generic phase ($t_0 = 0.37$, $\theta = 0.555$) the residual is finite and dominated by remap interpolation, as designed; see the caveat below. Figure 1 sweeps the shear phase across a full period: the residual drops by twelve orders of magnitude precisely at $t_0 = 0$ and $t_0 = T_{\rm sh}$ and is flat in between, confirming that the modular shear-phase arithmetic ($\theta = q\Omega\,[t \bmod T_{\rm sh}]$) has exactly the right period and that no error accumulates with time.

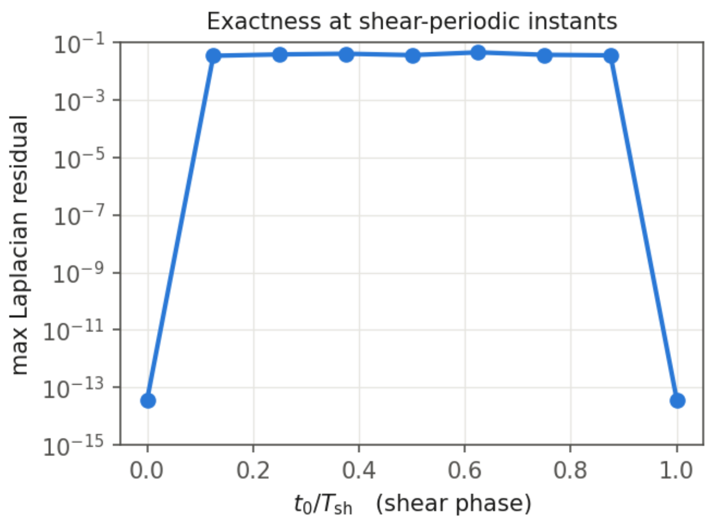

Figure 1: Laplacian residual versus shear phase over one shear period (64^3, `modes` profile, `ppmx` remap). The solve is exact to machine precision exactly when the box is strictly periodic.

5.3 Shearing-wave solutions: order-of-accuracy

For a single rolled-frame mode (n_1, n_2, n_3) the exact discrete solution is known analytically: the same slanted wave with amplitude $4\pi G\,\delta\rho/D$, D evaluated with the shifted k_x (Eq. 5). Figure 2 shows the pointwise potential error at $\theta = 0.555$. Measured orders per resolution doubling: `ppmx` 3.08, 3.02; `plm` 2.02, 2.01; `dc` 1.94, 2.05 — precisely the remap interpolation orders. Figure 3 shows the slanted wave and the solved potential (correlation coefficient -1.0 to 10^{-11}: potential wells align with density crests).

A caveat on metrics (important when re-validating). Under shear, the *Laplacian residual* of the returned potential is *not* a clean convergence metric: differencing divides the $O(\Delta y^p)$ remap interpolation error by Δ^2, so the residual improves only at roughly first order (and the max-norm can stall at under-resolved sharp features, where the `ppmx` limiter locally reduces order). This is intrinsic to the method — with spectral interpolation the residual would be exactly zero — and is why the pointwise-Φ comparison of Fig. 2 is the correct order-of-accuracy test. The printed `max_rel_int`/`l2_rel_int` norms exclude boundary layers whose ghosts are themselves interpolated (shear x_1 layers, extrapolated open-BC x_3 layers).

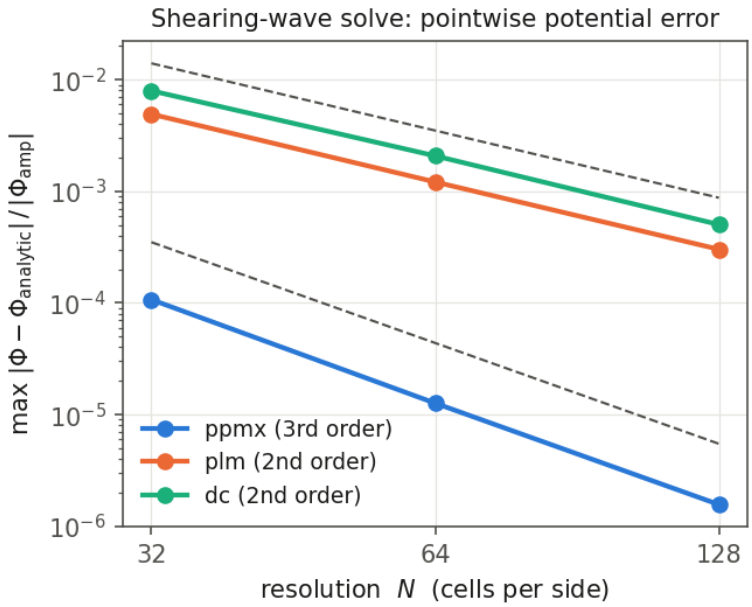

Figure 2: Maximum pointwise error of Φ against the analytic discrete shearing-wave solution, mode $(1, 1, 1)$, $q\Omega t_s = 0.555$, versus resolution, for the three remap orders. Dashed guides: N^{-2}, N^{-3}.

5.4 Open vertical boundary: slab potential

A horizontally uniform $\rho = \rho_0 \operatorname{sech}^2(z/H)$ slab with $H = L_z/20$ at 64^3, `vert_bc=open`: interior residual `4.2e-14` (machine precision — for this profile only the $k_\perp = 0$ column is active, where the two-solve weights are exact), and the $\Phi(z)$ profile matches the analytic infinite-slab potential $4\pi G\rho_0 H^2 \ln\cosh(z/H)$ to 0.15% (Fig. 4); the residual peaks at the slab center (finite-difference discretization of the sharp maximum) and is limited by the truncated sech^2 wings at the box scale.

5.5 End-to-end evolution: Jeans wave, FFT vs. multigrid

The existing Jeans-wave test (`pgen_name=gravity`) evolved to $t = 1$ (321 cycles, RK2 $\Rightarrow$ 642 Poisson solves, 128 meshblocks of 8^3):

	mode amplitude at t=1	wall time	zone-cycles/s
FFT (`jeans_fft.athinput`)	$-7.395719378 \times 10^{-7}$	6.6 s	3.2×10^6
Multigrid (`jeans_wave.athinput`)	$-7.395719385 \times 10^{-7}$	22.9 s	0.9×10^6

Agreement to nine significant digits (the difference is the multigrid iteration residual; the FFT solve is exact), with a 3.5× end-to-end speedup on this serial-CPU build. This exercises the full loop: per-stage solve, source-term coupling, and boundary/ghost handling during evolution.

5.6 Swing amplification: gravity coupled to the dynamics

All tests above check the Poisson solve in isolation. This one closes the loop: solver → source term → hydrodynamics, in a shearing frame.

A leading shearing wave ($k_{x0} < 0$, $k_y > 0$, $k_z = 0$) of small amplitude ($A = 10^{-4}$) is wound up by the background shear, $k_x(t) = k_{x0} + q\Omega k_y t$; as it swings through $k_x = 0$ self-gravity transiently

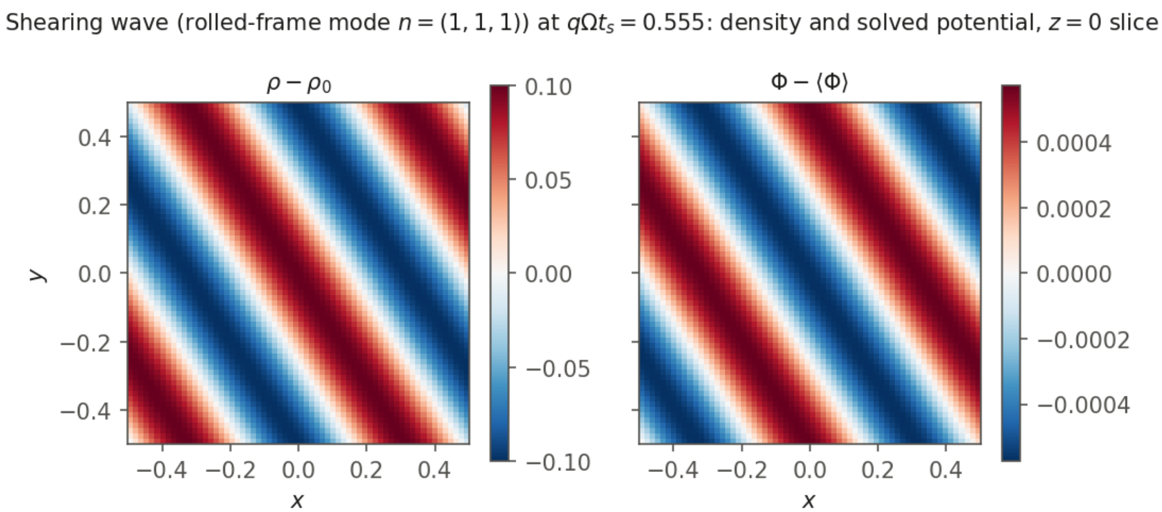

Figure 3: $z = 0$ slice of the shearing-wave test at 64^3: density perturbation (left) and solved potential (right). The wavefronts are slanted by the accumulated shear; Φ is exactly anticorrelated with $\delta\rho$.

amplifies it (Goldreich & Lynden-Bell 1965; Julian & Toomre 1966; Toomre 1981). Linearizing the isothermal shearing-box equations for one shearing wave, with $\delta = \delta\rho/\rho_0$ and $w_{x,y} = i\hat{v}_{x,y}$ (so $v = w \sin \mathbf{k}\cdot\mathbf{x}$ while $\delta \propto \cos$):

$$\delta' = -(k_x w_x + k_y w_y), \quad w_x' = 2\Omega w_y + k_x S, \quad w_y' = -(2-q)\Omega w_x + k_y S, \quad S = \delta\Big(c_s^2 + \frac{4\pi G\rho_0}{D}\Big), \tag{9}$$

where D is the solver's own discrete eigenvalue (Eq. 2, $\to -k^2$ in the continuum). Eq. (9) is derived from the shearing-box equations in Appendix A; setting $k_y = 0$ there recovers the Jeans dispersion relation $\delta'' = -(k^2 c_s^2 - 4\pi G\rho_0)\delta$, one of three checks collected in A.6.

Parameters are chosen in the epicyclic-resonant regime where swing is strong: $L_x = L_y = 2\pi$, $n_{wy} = 1$ so $k_y = 1 = \kappa$ (with $\kappa^2 = 2(2-q)\Omega^2 = 1$ at $q = 1.5$), and $4\pi G\rho_0 = 1.9$, just below the marginal value $k_y^2 c_s^2 + \kappa^2 = 2$ — so the swinging wave is strongly amplified while every box mode remains Jeans-stable. Starting from $n_{wx} = -6$ the swing occurs at $t_{\text{swing}} = -n_{wx}/(q\, n_{wy}) = 4$, and the run covers $t \in [0, 8]$ (12 shear periods). The `swing` problem generator projects the solution onto the *instantaneous* wavevector each history dump, giving δ, w_x, w_y directly.

Results (Fig. 5): the simulation tracks the linear solution through the entire swing, reaching a peak amplification of 3.51 against the predicted 3.60, with relative L_2 differences of 9×10^{-3} (δ), 8×10^{-3} (w_x) and 1.3×10^{-2} (w_y) over the whole run at 128^2. Two sharper diagnostics:

- The $k_x = 0$ crossing recorded in the history file occurs at $t = 4.005$ against the predicted 4.000 — the shear-phase bookkeeping is right after twelve shear periods.
- The *sine* projection of the density stays at $|\delta_{\sin}|/|\delta|_{\max} \sim 10^{-13}$ throughout. Linear theory says the wave stays exactly in phase with $\cos(\mathbf{k}(t)\cdot\mathbf{x})$; the simulation reproduces that phase locking to machine precision, which is a far more stringent statement than the amplitude agreement.

The residual $\sim$1% difference is hydrodynamic truncation error, not a gravity error. Table 1 makes that quantitative by repeating the resolution study with self-gravity disabled, comparing each run against its own linear theory ($4\pi G = 1.9$ and $4\pi G = 0$ respectively). Two things follow:

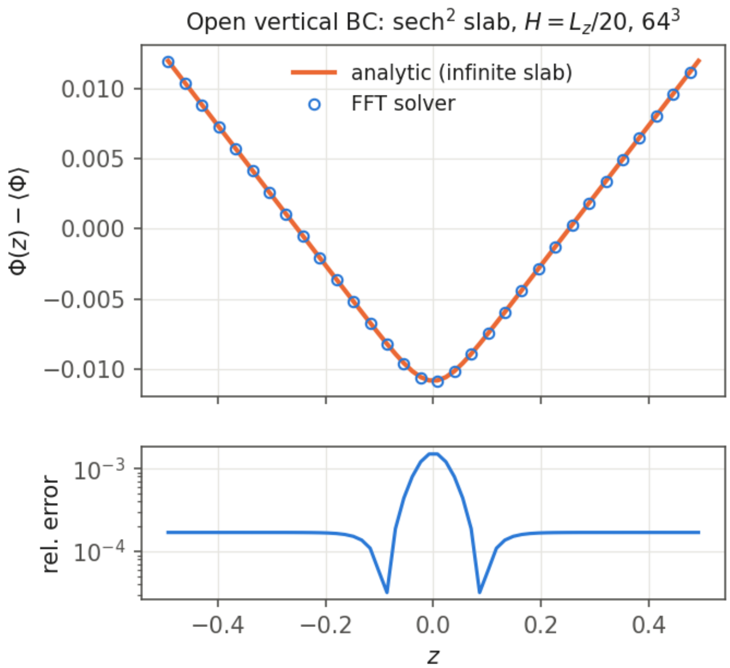

Figure 4: Open-BC slab test: solved $\Phi(z)$ (circles) versus the analytic infinite-slab potential (line), both zero-mean; lower panel: pointwise relative error.

- The peak amplification converges cleanly onto the predicted value, $2.64 \to 3.51 \to 3.62$ against 3.605 (a 0.4% difference at 256^2), while the control converges onto its own, much smaller, gravity-free value (1.80 vs. 1.49 predicted — the control is *not* amplified).
- At 256^2 the two L_2 errors are equal to within 3% (4.10×10^{-3} with gravity, 3.99×10^{-3} without). An error that is the same size whether or not the Poisson solver is running cannot originate in the Poisson solver: it is the PLM/HLLE/RK2 discretization of the hydrodynamics, which the linear ODE does not model. (The gravity-on sequence is non-monotonic in apparent order only because its 64^2 point is badly under-resolved — the peak amplitude is off by 27% there.)

Table 1: Swing solution versus linear theory (relative L_2 over $t \in [0, 8]$, each run compared against theory with its own $4\pi G$).

	self-gravity ON		self-gravity OFF (control)	
resolution	peak δ/δ_0	L_2 error	peak δ/δ_0	L_2 error
$64^2 \times 16$	2.638	9.08×10^{-2}	1.275	9.78×10^{-2}
$128^2 \times 16$	3.511	8.97×10^{-3}	1.734	1.81×10^{-2}
$256^2 \times 16$	3.619	4.10×10^{-3}	1.801	3.99×10^{-3}
linear theory	3.605	—	1.490	—

5.7 Reproducing

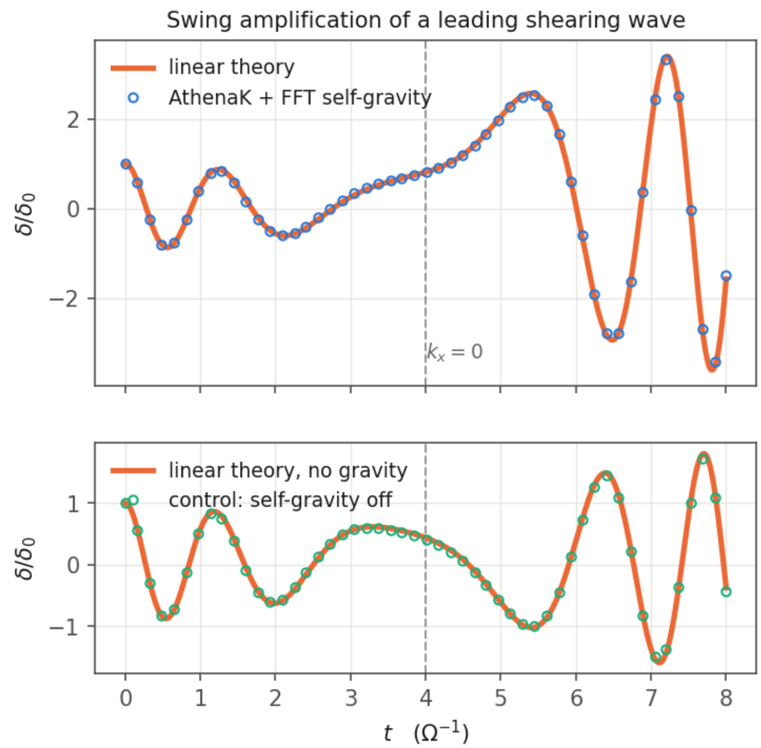

Figure 5: Swing amplification at $128^2 \times 16$. Top: density amplitude of the shearing wave versus time, simulation (circles) against the linear ODE (Eq. 9, line); dashed line marks $k_x = 0$. Bottom: the same run with self-gravity disabled, against the $4\pi G = 0$ theory — no amplification.

```
# periodic consistency
./build/src/athena -i inputs/tests/fft_poisson.athinput
# shear: generic phase / periodic instant
./build/src/athena -i inputs/tests/fft_poisson_shear.athinput
./build/src/athena -i inputs/tests/fft_poisson_shear.athinput \
                   problem/time0=0.6666666666666667
# shearing-wave convergence (vary N, gravity/fft_remap, problem/n1..n3)
./build/src/athena -i inputs/tests/fft_poisson_shear.athinput \
    problem/profile=shwave gravity/fft_remap=ppmx \
    mesh/nx1=64 mesh/nx2=64 mesh/nx3=64 \
    meshblock/nx1=64 meshblock/nx2=64 meshblock/nx3=64
# open-BC slab
./build/src/athena -i inputs/tests/fft_poisson_open.athinput
# Jeans evolution
./build/src/athena -i inputs/tests/jeans_fft.athinput
# swing amplification (writes SwingSG.user.hst); control run without gravity:
./build/src/athena -i inputs/tests/swing_selfgrav.athinput
./build/src/athena -i inputs/tests/swing_selfgrav.athinput \
    job/basename=SwingNoG128 hydro_srcterms/self_gravity=false
```

Each run writes `bin/*.bin` snapshots (`hydro_w`, `grav_phi`) that the Julia notebook reads. Figures in this document are regenerated by `validation/figures/make_figures.py`.

6 Next steps

1. Multi-rank support (`MPI_Allgatherv` at the seam documented in Sec. 3), lifting the single-rank restriction.

2. Python regression harness under `tst/test_suite/fft_gravity/` mirroring the multigrid suite, so these checks run in CI.

3. The FFT-root + multigrid-refinement hybrid for the AMR goal: the FFT solve on the uniform root grid supplies both the global solution and, by prolongation, Dirichlet boundary data for multigrid on refined interior patches — which never touch the shear-periodic boundary.

4. Physics extensions once the above land: star-particle deposition into the density field (skipped in the Athena-C shearing branch, see Sec. 3), and stratified shearing boxes combining `vert_bc=open` with vertical gravity.

A Derivation of the linearized shearing-wave system, Eq. (9)

This appendix derives Eq. (9) — the reference solution against which the swing test of Sec. 5.6 is compared — from the shearing-box equations. Three checkpoints for verifying the algebra are collected in A.6.

A.1 Shearing-box equations and the background state

The shearing box is a local Cartesian frame co-rotating at radius R_0 with angular velocity $\mathbf{\Omega} = \Omega\hat{z}$; coordinates (x, y, z) are (radial, azimuthal, vertical) with $x = R - R_0$. For an isothermal, unstratified gas ($P = c_s^2\rho$, no vertical external gravity):

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0, \qquad \frac{\partial\mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\frac{c_s^2}{\rho}\nabla\rho - \nabla\Phi \underbrace{-\,2\mathbf{\Omega}\times\mathbf{v}}_{\text{Coriolis}} + \underbrace{2q\Omega^2 x\,\hat{x}}_{\text{tidal}}, \tag{10}$$

with $\nabla^2\Phi = 4\pi G\rho$ and $q \equiv -\mathrm{d}\ln\Omega/\mathrm{d}\ln R$ ($q = 3/2$ Keplerian). The tidal term is the first-order expansion in x of the combined centrifugal and central-gravity forces; the Coriolis term written out is $-2\Omega\hat{z}\times\mathbf{v} = 2\Omega v_y\hat{x} - 2\Omega v_x\hat{y}$.

The background is uniform density in shear flow,

$$\rho = \rho_0, \qquad \mathbf{v}_0 = -q\Omega x\,\hat{y}, \qquad \Phi_0 = 0. \tag{11}$$

That this solves Eq. (10) is worth checking explicitly, because it fixes the sign convention for everything that follows: the x-momentum equation reduces to $0 = 2\Omega v_{0y} + 2q\Omega^2 x = -2q\Omega^2 x + 2q\Omega^2 x = 0$, i.e. the Coriolis force on the shear flow exactly cancels the tidal force. Setting $\Phi_0 = 0$ despite $\rho_0 \neq 0$ is the Jeans swindle; the solver enforces precisely this by zeroing the $\mathbf{k} = 0$ Fourier mode (Sec. 2.2).

A.2 Linearization

Write $\rho = \rho_0(1 + \delta)$, $\mathbf{v} = \mathbf{v}_0 + \mathbf{v}'$, $\Phi = \Phi'$, and define the background-advection operator

$$\frac{D}{Dt} \;\equiv\; \frac{\partial}{\partial t} + (\mathbf{v}_0\cdot\nabla) \;=\; \frac{\partial}{\partial t} - q\Omega x\frac{\partial}{\partial y}. \tag{12}$$

Keeping terms of first order in $(\delta, \mathbf{v}', \Phi')$, and using $\nabla \cdot \mathbf{v}_0 = \partial_y(-q\Omega x) = 0$:

$$\frac{D\delta}{Dt} = -\nabla \cdot \mathbf{v}', \tag{13}$$

$$\frac{Dv'_x}{Dt} = -c_s^2 \frac{\partial \delta}{\partial x} - \frac{\partial \Phi'}{\partial x} + 2\Omega v'_y, \tag{14}$$

$$\frac{Dv'_y}{Dt} = -c_s^2 \frac{\partial \delta}{\partial y} - \frac{\partial \Phi'}{\partial y} - (2-q)\Omega v'_x. \tag{15}$$

The only non-obvious term is the last one. The advection of the *background* azimuthal velocity by the radial perturbation contributes $(\mathbf{v}' \cdot \nabla) v_{0y} = v'_x\, \partial_x(-q\Omega x) = -q\Omega v'_x$ on the left-hand side; moving it right turns the bare Coriolis coefficient -2Ω into $-(2-q)\Omega$. This $(2-q)\Omega$ is the fingerprint of working with velocity *fluctuations* rather than total velocity, and it is exactly the coefficient appearing in AthenaK's `shearing_box_srcterms.cpp` — which is how the convention assumed here was confirmed against the code.

A.3 The shearing-wave ansatz

Because D/Dt carries an explicit x-dependent coefficient, plane waves of fixed wavevector are not solutions. Try instead a wave whose radial wavenumber depends on time,

$$f(\mathbf{x}, t) = \hat{f}(t)\, e^{i[k_x(t)x + k_y y + k_z z]} \quad \Longrightarrow \quad \frac{Df}{Dt} = \left[\frac{d\hat{f}}{dt} + i\Big(\frac{dk_x}{dt} - q\Omega k_y \Big) x\, \hat{f} \right] e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{16}$$

The residual x-dependence cancels if and only if

$$\frac{dk_x}{dt} = q\Omega k_y \qquad \Longrightarrow \qquad k_x(t) = k_{x0} + q\Omega k_y\, t, \tag{17}$$

after which $D/Dt \to d/dt$ acting on the amplitudes. The shearing wave is therefore *forced* by the structure of the operator rather than assumed. Eq. (17) is the same relation that appears in the solver's k-space kernel (Eq. 5), in the `swing` history projection, and in AthenaK's pre-existing `shwave.cpp`.

A.4 Amplitude equations and the Poisson closure

Substituting $\nabla \to i\mathbf{k}(t)$ and specializing to $k_z = 0$ (so that $\hat{v}_z$ decouples and remains zero):

$$\frac{d\hat{\delta}}{dt} = -i(k_x \hat{v}_x + k_y \hat{v}_y), \quad \frac{d\hat{v}_x}{dt} = 2\Omega \hat{v}_y - ik_x S, \quad \frac{d\hat{v}_y}{dt} = -(2-q)\Omega \hat{v}_x - ik_y S, \tag{18}$$

with the Poisson closure

$$S \equiv c_s^2 \hat{\delta} + \hat{\Phi}, \qquad \hat{\Phi} = \frac{4\pi G \rho_0 \hat{\delta}}{D} \quad \Longrightarrow \quad S = \hat{\delta}\left(c_s^2 + \frac{4\pi G\rho_0}{D} \right). \tag{19}$$

Here D is deliberately the *discrete* eigenvalue of Eq. (2) rather than its continuum limit $-k^2$, so that the reference solution inverts the same operator the solver does; the difference is $O((k\Delta x)^2)$ and is not negligible at the resolutions used. Note $D < 0$ for every non-zero mode, so self-gravity always *reduces* the restoring force, as it must.

One geometric caveat: with $k_z = 0$ in a three-dimensional periodic box the closure is $\hat{\Phi} = -4\pi G\rho_0 \hat{\delta}/k_\perp^2$, which is *not* the razor-thin sheet result $\hat{\Phi} = -2\pi G\hat{\Sigma}/|k_\perp|$ used in much of the swing-amplification literature. Both are correct in their own geometry; the test here is a 3D box with a vertically uniform perturbation, so the former applies.

A.5 Real form

Finally, substituting $w_x \equiv i\hat{v}_x$, $w_y \equiv i\hat{v}_y$ removes every explicit i and yields Eq. (9):

$$\delta' = -(k_x w_x + k_y w_y), \qquad w_x' = 2\Omega w_y + k_x S, \qquad w_y' = -(2-q)\Omega w_x + k_y S. \tag{20}$$

Because the coefficients are real, real initial data stays real for all time, which has a directly testable consequence: the density perturbation remains exactly in phase with $\cos(\mathbf{k}(t)\cdot\mathbf{x})$ and the velocities with $\sin(\mathbf{k}(t)\cdot\mathbf{x})$, with no phase drift. This is why the `swing` history function projects density onto the cosine and velocity onto the sine of the instantaneous phase, and why the measured sine component of the density (Sec. 5.6, $\sim 10^{-13}$) is such a stringent check.

A.6 Checks on the algebra

Three limits verify the system independently:

1. **Jeans limit** ($k_y = 0$, $\Omega = 0$): eliminating w_x gives $\delta'' = -(k^2 c_s^2 - 4\pi G\rho_0)\delta$, the Jeans dispersion relation $\omega^2 = k^2 c_s^2 - 4\pi G\rho_0$. The companion notebook integrates the ODE in this limit and reproduces the analytic $\cos(\omega t)$ / $\cosh(|\omega| t)$ solutions to six decimal places in both the stable and unstable regimes.
2. **Epicyclic limit** ($S = 0$): $w_x'' = -\kappa^2 w_x$ with $\kappa^2 = 2(2-q)\Omega^2$, the epicyclic frequency ($\kappa = \Omega$ for $q = 3/2$). Obtaining $\kappa^2 = 4\Omega^2$ instead means the $(\mathbf{v}'\cdot\nabla)v_{0y}$ term in Eq. (15) was dropped.
3. **Marginal stability at swing** ($k_x = 0$): the tightly-wound dispersion relation $\omega^2 = k_y^2 c_s^2 - 4\pi G\rho_0 + \kappa^2$ sets the parameter choice of Sec. 5.6 — $4\pi G\rho_0 = 1.9$ against the marginal value $k_y^2 c_s^2 + \kappa^2 = 2$.